Face posets, local weight matroids and Betti numbers of torus actions

David A. Mikheev

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Abstract

Let a compact torus act on a closed manifold with isolated fixed points. Its face poset $S(X)$ is covered by the lattices of flats of the linear matroids of tangent weights at the fixed points. We study how much of the cohomology these combinatorial data see. First, the local equivariant cohomology $H^*_T(X, X \setminus \mathrm{pt})$, embedded in $H^*_T(\mathrm{pt})$ by restriction, is the principal ideal generated by the product of the tangent weights; it therefore determines the weights up to scalars, with multiplicities, and hence the local lattice $S(X)_{\geqslant \mathrm{pt}}$. Second, the local weight matroids do not determine the equivariant cohomology: two connected regular semisimple Hessenberg varieties in $\mathrm{Fl}(\mathbb{C}^5)$, with actions of complexity one, have isomorphic weight matroids at all fixed points but different Betti numbers. Third, the global face poset does determine the Betti numbers for every regular semisimple Hessenberg variety of type A, full or partial, and for the isospectral manifolds of Ayzenberg and Buchstaber. The reason is rigidity: the cluster-permutohedron Cl_Γ determines the graph Γ up to isomorphism, by an elementary argument that recovers the vertices of Γ from stars of edges.

1 Introduction

Throughout, a *torus action* is an effective action of a compact torus $T = T^k$ on a closed smooth manifold $X = X^{2n}$ with a finite nonempty set of fixed points. Toric topology studies such actions of complexity $n - k$ zero by combinatorial means [8]. Buchstaber and Terzić extended these methods to positive complexity [9, 10], and Ayzenberg, Masuda and their coauthors made the *face poset* $S(X)$ of the action the central combinatorial object [5, 6, 4]. Its elements are the closures of the components of the orbit type strata of X/T that contain a fixed point, ordered by inclusion. By a theorem of Ayzenberg and Cherepanov [5], $S(X)$ is *locally geometric*: for a fixed point p, the faces containing p form the lattice of flats of the linear matroid of the tangent weights at p.

This paper asks what these combinatorial data know about the cohomology of X. We prove three results, one local and two global.

Theorem A (Theorem 3.3)**.** *Let* pt *be a fixed point with tangent weights* $\alpha_1, \dots, \alpha_n \in H^2(BT; \mathbb{Q})$. *Restriction identifies* $H^*_T(X, X \setminus \mathrm{pt})$ *with the principal ideal* $(\alpha_1 \cdots \alpha_n)$ *of* $H^*_T(\mathrm{pt}) = H^*(BT; \mathbb{Q})$. *This ideal determines the multiset of weights up to nonzero scalars, hence the linear matroid of weights at* pt *with its parallel classes, and the lattice* $S(X)_{\geqslant \mathrm{pt}}$.

The point of Theorem A is that no independence assumption on the weights is needed. The familiar GKM argument computes the ideal as an intersection $\bigcap_j (\alpha_j)$, which equals the product only when the weights are pairwise independent and loses multiplicities otherwise. We use the Thom isomorphism and unique factorization instead.

The local data are far from determining the global cohomology. The Chow ring [2] and the Orlik–Solomon algebra [19] of a matroid depend only on the matroid, so the following shows that neither of them, taken at all fixed points, determines $H^*_T(X)$.

Theorem B (Theorem 4.3)**.** *Let* $M = \mathrm{diag}(1,2,3,4,5)$*. The regular semisimple Hessenberg varieties* $X_1 = \mathrm{Hess}(M,(2,3,5,5,5))$ *and* $X_2 = \mathrm{Hess}(M,(2,4,4,5,5))$ *in* $\mathrm{Fl}(\mathbb{C}^5)$ *are connected GKM manifolds with the same* 120 *fixed points and torus actions of complexity one. At every fixed point of either variety the weight matroid is* $U_{2,3}\oplus U_{1,1}\oplus U_{1,1}$*. Their Poincaré polynomials are* $1+17q+42q^2+42q^3+17q^4+q^5$ *and* $1+12q+47q^2+47q^3+12q^4+q^5$*, where* q *has degree* 2*.*

The global face posets of X_1 and X_2 are different. This leaves the question, raised in [16], whether the face poset determines the Betti numbers of an equivariantly formal action. In complexity zero it does: for a quasitoric manifold the Betti numbers are the h-numbers of the orbit polytope [8]. Our third result answers the question for the main family of examples in positive complexity.

Theorem C (Theorems 5.4 and 5.8, Corollary 5.6). (1) For simple graphs Γ,Γ', an isomorphism of cluster-permutohedra $\mathrm{Cl}_\Gamma \cong \mathrm{Cl}_{\Gamma'}$ implies $\Gamma \cong \Gamma'$.

(2) Let X and X' each be an isospectral manifold $M_{\Gamma,\lambda}$ or a regular semisimple Hessenberg variety $\mathrm{Hess}(M,h)$ of type A, with Γ, resp. Γ_h, connected. If $S(X) \cong S(X')$, then X and X' have the same Betti numbers and isomorphic equivariant cohomology rings.

(3) The same conclusion about Betti numbers holds for partial regular semisimple Hessenberg varieties, including all partial flag varieties.

The cluster-permutohedron Cl_Γ is the face poset of the isospectral manifold $M_{\Gamma,\lambda}$, and through the twin construction of Ayzenberg and Buchstaber also of $\mathrm{Hess}(M,h)$ when $\Gamma = \Gamma_h$ [3, 4]. Part (1) is purely combinatorial. Counting the minimal elements below joins of pairs of atoms tells which edges of Γ meet, i.e. recovers the line graph, and ranks of joins tell which triples of edges form triangles. These two data determine Γ by an elementary star argument (Proposition 5.3). Unlike Whitney's line graph theorem [21, 15], it has no exceptions. For partial flags we combine the same argument with a fibration over the partial Hessenberg variety.

Theorem C does not give a formula for the Betti numbers in terms of the poset. It shows that in these families the face poset remembers the whole graph, and the graph remembers the manifold. A counterexample to the general question, if one exists, must therefore come from outside type A.

Computations and formalization. All computations cited in this paper were done with independent programs, which are available with their output at `https://github.com/davidmiheev/matroid_toric`. The Betti numbers in Theorem B were computed both from the Białynicki-Birula formula and, independently, from the GKM presentation of H^*_T in Macaulay2 [14]. Part (1) of Theorem C was confirmed by computing exact poset isomorphism classes, with nauty's Traces [17], for all 208 graphs with at most 6 vertices. Before the proof, Theorem C(2) and (3) were verified by direct computation for all full and partial Hessenberg varieties with $n \leqslant 7$ (714 and 2 940 cases). Part (1) of Theorem C (Theorem 5.4, including Proposition 5.3) is formally verified in Lean 4 with Mathlib, in the same repository, with the cluster-permutohedron encoded as maps from vertices to label sets ordered pointwise. So is the algebraic step of Theorem A, Lemma 3.2(2), together with the lattice of flats of a matroid, Whitney's Boolean expansion of its characteristic polynomial and Philip Hall's theorem.

Conventions. Cohomology has rational coefficients. We identify $H^*(BT) = \mathrm{Sym}(t^*_{\mathbb{Q}}) \cong \mathbb{Q}[t_1,\dots,t_k]$ with $\deg t_i = 2$ and $H^2(BT) = \mathrm{Hom}(T,S^1)\otimes\mathbb{Q}$. For a polynomial in q we write q for the class of degree 2, so $P_X(q) = \sum_j \beta_{2j}(X)q^j$ when $H^{\mathrm{odd}}(X) = 0$.

2 Preliminaries

2.1 Weights, face posets and GKM manifolds

Let $p \in X^T$. The tangent representation decomposes as $T_pX \cong \bigoplus_{j=1}^{n} V(\alpha_{p,j})$, where $V(\alpha) = \mathbb{C}$ with $t \cdot z = \alpha(t)z$. Without an invariant complex structure the weights $\alpha_{p,j} \in \mathrm{Hom}(T, S^1)$ are defined up to sign, which does not affect linear dependence. The point p is isolated if and only if no weight vanishes. For an oriented T-vector bundle, the equivariant Euler class of T_pX at a fixed point is $e_T(T_pX) = \pm \prod_j \alpha_{p,j} \in H^{2n}(BT)$.

Let $X_j \subset X$ be the union of the orbits of dimension at most j. A *face* of rank j is the closure of a connected component of $X_j/T \setminus X_{j-1}/T$ that contains a fixed point, and $S(X)$ is the set of faces ordered by inclusion.

Theorem 2.1 (Ayzenberg–Cherepanov [5, Thm. 1])**.** *For every $p \in X^T$, the poset $S(X)_{\geqslant p}$ of faces containing p is isomorphic to the lattice of flats of the linear matroid of the weights $\alpha_{p,1}, \dots, \alpha_{p,n} \in \mathrm{Hom}(T, S^1) \otimes \mathbb{Q}$. In particular $S(X)$ is graded, and each $S(X)_{\geqslant p}$ is a geometric lattice.*

The action is *equivariantly formal* if the Serre spectral sequence of $X_T \to BT$ degenerates; for X^T finite and nonempty this holds if and only if $H^{\mathrm{odd}}(X) = 0$, and then $H^*_T(X) \cong H^*(X) \otimes H^*(BT)$ as graded $H^*(BT)$-modules. A *GKM manifold* is an equivariantly formal X whose weights at each fixed point are pairwise linearly independent. Its fixed points and invariant 2-spheres form the *GKM graph*, whose edge pq is labelled by the weight α_{pq} along the sphere, and by the GKM theorem [13, 11]

$$H^*_T(X) \cong \left\{ (f_p) \in \textstyle\bigoplus_{p \in X^T} H^*(BT) \;\middle|\; f_p - f_q \in (\alpha_{pq}) \text{ for every edge } pq \right\}.$$

2.2 Hessenberg varieties

A *Hessenberg function* is a non-decreasing map $h\colon [n] \to [n]$ with $h(j) \geqslant j$. For $M = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ with pairwise distinct real λ_i, the *regular semisimple Hessenberg variety* is

$$\mathrm{Hess}(M, h) = \{ V_\bullet \in \mathrm{Fl}(\mathbb{C}^n) \mid MV_i \subset V_{h(i)} \text{ for all } i \}.$$

It is a smooth projective variety of complex dimension $\sum_j (h(j) - j)$ with $H^{\mathrm{odd}} = 0$ [12, 1], invariant under the diagonal torus $T = T^n$. The fixed points are the $n!$ coordinate flags, indexed by $w \in S_n$, and the tangent weights at w are $t_{w(i)} - t_{w(j)}$ for $j < i \leqslant h(j)$ [12, 20]. Let Γ_h be the graph on $[n]$ with edges $\{j, i\}$, $j < i \leqslant h(j)$. Up to isomorphism, the graphs Γ_h are exactly the indifference (unit interval) graphs [3]. By [12],

$$P_{\mathrm{Hess}(M,h)}(q) = \sum_{w \in S_n} q^{\ell_h(w)}, \qquad \ell_h(w) = \#\{(j, i) \mid j < i \leqslant h(j),\ w(j) > w(i)\}. \tag{2.1}$$

Let $\bar{k} = (k_1, \dots, k_s)$ be a composition of n, $K_b = k_1 + \cdots + k_b$, and h a Hessenberg function on $[s]$. The *partial Hessenberg variety* $\mathrm{Hess}_{\bar{k}}(M, h)$ is the set of partial flags $V_{K_1} \subset \cdots \subset V_{K_s} = \mathbb{C}^n$ with $MV_{K_a} \subset V_{K_{h(a)}}$ for all a. It is smooth (Lemma 5.10). Its fixed points are the ordered set partitions $J = (J_1, \dots, J_s)$ of $[n]$ with $|J_a| = k_a$, and its tangent weights at J are $t_y - t_x$ for $x \in J_a$, $y \in J_b$, $a < b \leqslant h(a)$ [4, §5]. In [4] this variety is denoted $M'_{\bar{k},\Gamma_h,\lambda}$ and is defined by the conditions $ML_a \subset \bigoplus_b L_b$, the sum over $b = a$ and the neighbours b of a in Γ_h, on the orthogonal pieces $L_a = V_{K_a} \cap V^{\perp}_{K_{a-1}}$. Since M is real and diagonal, hence Hermitian, $ML_a \perp L_b$ if and only if $ML_b \perp L_a$, and the two definitions agree. For a generic one-parameter subgroup, the Białynicki-Birula decomposition gives

$$P_{\mathrm{Hess}_{\bar{k}}(M,h)}(q) = \sum_J q^{\mathrm{ind}(J)}, \qquad \mathrm{ind}(J) = \#\{(x, y) \mid x \in J_a,\ y \in J_b,\ a < b \leqslant h(a),\ x > y\}, \tag{2.2}$$

which reduces to (2.1) for $\bar{k} = (1, \dots, 1)$. The *inflation* $\Gamma_h(\bar{k})$ is the graph obtained from Γ_h by replacing each vertex a by k_a pairwise non-adjacent copies, copies of a and b being adjacent iff ab is an edge of Γ_h. At the fixed point J, identifying $x \in J_a$ with a copy of a, the tangent weights are $t_x - t_y$ over the edges xy of $\Gamma_h(\bar{k})$, so the local weight matroid is the graphic matroid $M(\Gamma_h(\bar{k}))$.

2.3 Isospectral manifolds and cluster-permutohedra

Let Γ be a connected graph on $[n]$ and λ a generic spectrum. The *isospectral manifold* $M_{\Gamma,\lambda}$ is the space of Hermitian matrices A with spectrum λ and $a_{ij} = 0$ whenever $\{i, j\} \notin E(\Gamma)$, with T^n acting by conjugation [4]. For $\Gamma = K_n$ it is $\mathrm{Fl}(\mathbb{C}^n)$.

Definition 2.2 ([4, Def. 2.14])**.** Let Γ be a simple graph on $[n]$. A *clustering* is a partition $C = \{V_1, \dots, V_m\}$ of $[n]$ into blocks inducing connected subgraphs; ordered by refinement, the clusterings form the lattice of flats $\mathrm{Flats}(\Gamma)$ of the graphic matroid. An *assignment* for C is a family of label sets $L_1, \dots, L_m$ partitioning $[n]$ with $|L_i| = |V_i|$. The *cluster-permutohedron* Cl_Γ is the set of pairs (clustering, assignment), where $(C', A') \leqslant (C, A)$ if C' refines C and the label set of each block of C is the union of the label sets of the blocks of C' it contains. It is graded by $\mathrm{rk}(C, A) = n - |C|$.

Theorem 2.3 (Ayzenberg–Buchstaber [4, Thm. 1, Cor. 2.43], [3])**.** *For connected Γ and generic λ, $S(M_{\Gamma,\lambda}) \cong \mathrm{Cl}_\Gamma$. If Γ_h is connected, then $\mathrm{Hess}(M, h)$ and $M_{\Gamma_h,\lambda}$ are twins: they have homeomorphic orbit spaces by a homeomorphism preserving orbit dimensions, equivariant cohomology rings that are isomorphic as graded rings, and equal Betti numbers [3, Thm. 3.11]. In particular $S(\mathrm{Hess}(M, h)) \cong \mathrm{Cl}_{\Gamma_h}$. The isomorphism of equivariant cohomology is not one of $H^*(BT)$-algebras in general: the weights of $M_{\Gamma_h,\lambda}$ at the fixed point w are $t_i - t_j$, those of $\mathrm{Hess}(M, h)$ are $t_{w(i)} - t_{w(j)}$. The face poset of a partial Hessenberg variety is described in Lemma 5.7, following [4, Thm. 2].*

3 Local equivariant cohomology

Let $V \cong \bigoplus_{j=1}^{n} V(\alpha_j)$ be a representation of T with all $\alpha_j \neq 0$.

Proposition 3.1. *Restriction to 0 is an injective map $H^*_T(V, V \setminus 0) \to H^*_T(0) = H^*(BT)$ with image the principal ideal $(\alpha_1 \cdots \alpha_n)$.*

Proof. Choose an invariant complex structure on each $V(\alpha_j)$, which orients V. The group $H^*_T(V, V \setminus 0)$ is the cohomology of the Thom space of the oriented bundle $V_T = ET \times_T V \to BT$. By the Thom isomorphism [18] it is free of rank one over $H^*(BT)$ on the Thom class τ, and τ restricts to the Euler class $e(V_T) = \prod_j \alpha_j$ in $H^*_T(V) = H^*_T(0)$. Restriction to 0 is therefore multiplication by $\prod_j \alpha_j$, which is injective since $H^*(BT)$ is a domain. □

Lemma 3.2. *Let k be a field and $R = k[t_1, \dots, t_m]$.*

(1) *Every nonzero linear form is irreducible, hence prime, in R.*

(2) *If $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_{n'}$ are nonzero linear forms with $(\alpha_1 \cdots \alpha_n) = (\beta_1 \cdots \beta_{n'})$, then $n = n'$ and there are $\sigma \in S_n$ and $c_j \in k^\times$ with $\beta_{\sigma(j)} = c_j \alpha_j$.*

Proof. (1) A factorization of a linear form has a factor of degree 0, a unit. (2) Equal principal ideals of a domain have associate generators, and both products are factorizations into primes by (1); uniqueness of factorization in R gives the claim. □

Theorem 3.3. *Let $pt \in X^T$ be an isolated fixed point with tangent weights $\alpha_1, \dots, \alpha_n$.*

(1) *The composition $H^*_T(X, X \setminus pt) \to H^*_T(X) \to H^*_T(pt)$ is injective, with image $(\alpha_1 \cdots \alpha_n) = (e_T(T_{pt}X))$.*

(2) *This ideal determines the multiset* $\{\alpha_j\}$ *up to nonzero scalars. Hence it determines the linear matroid of weights at* pt*, including parallel elements, and by Theorem 2.1 the lattice* $S(X)_{\geqslant \mathrm{pt}}$.

Proof. By the slice theorem [7], pt has an invariant closed neighbourhood D equivariantly diffeomorphic to the unit disc of $V = T_{\mathrm{pt}}X$. Excision of $X \setminus \operatorname{int} D$ and radial deformation give $H^*_T(X, X \setminus \mathrm{pt}) \cong H^*_T(D, D \setminus \mathrm{pt}) \cong H^*_T(V, V \setminus 0)$, compatibly with restriction to pt. Part (1) follows from Proposition 3.1, and part (2) from Lemma 3.2, since rescaling does not change linear dependence. □

Remark 3.4. As an abstract module, $H^*_T(X, X \setminus \mathrm{pt})$ is free of rank one and carries no information; everything lies in its embedding into $H^*_T(\mathrm{pt})$. For a GKM manifold the image can also be computed from the GKM theorem, as $\bigcap_j(\alpha_j)$. Without pairwise independence this intersection differs from the product: for $\alpha_1 = \alpha_2$ it is (α_1), not (α_1^2). The Thom isomorphism argument needs no such assumption, which is why Theorem 3.3 sees multiplicities.

4 Local weight matroids do not determine equivariant cohomology

Let $M = \operatorname{diag}(1, 2, 3, 4, 5)$, $T = T^5$, and

$$h_1 = (2, 3, 5, 5, 5), \quad h_2 = (2, 4, 4, 5, 5), \quad X_1 = \operatorname{Hess}(M, h_1), \quad X_2 = \operatorname{Hess}(M, h_2).$$

So $X_1 = \{MV_1 \subset V_2,\ MV_2 \subset V_3\}$ and $X_2 = \{MV_1 \subset V_2,\ MV_2 \subset V_4,\ MV_3 \subset V_4\}$. The graph Γ_{h_1} has edges $12, 23, 34, 35, 45$: a triangle $\{3, 4, 5\}$ with the path $1-2-3$ attached. The graph Γ_{h_2} has edges $12, 23, 24, 34, 45$: a triangle $\{2, 3, 4\}$ with two pendant edges. They are not isomorphic.

Lemma 4.1. *At every fixed point* w *of* $\operatorname{Hess}(M, h)$*, the weight matroid is isomorphic to the graphic matroid* $M(\Gamma_h)$.

Proof. The weights at w are $e_{w(i)} - e_{w(j)}$ over the edges $\{j, i\}$ of Γ_h. The linear automorphism $e_a \mapsto e_{w^{-1}(a)}$ carries them to $e_i - e_j$, which realize $M(\Gamma_h)$. □

Lemma 4.2. $M(\Gamma_{h_1}) \cong M(\Gamma_{h_2}) \cong U_{2,3} \oplus U_{1,1} \oplus U_{1,1}$.

Proof. In each graph the only cycle is the triangle and the other two edges are bridges. □

Theorem 4.3. (1) X_1 *and* X_2 *are connected smooth projective varieties of complex dimension* 5 *with* $H^{\mathrm{odd}} = 0$*. The torus* T *acts on both with the same* 120 *fixed points, as GKM actions whose effective quotient* $T/\Delta(S^1) \cong T^4$ *acts with complexity one.*

(2) *For all* $p_1 \in X_1^T$ *and* $p_2 \in X_2^T$ *the weight matroids are isomorphic to* $U_{2,3} \oplus U_{1,1} \oplus U_{1,1}$*. Hence* $S(X_1)_{\geqslant p_1} \cong S(X_2)_{\geqslant p_2}$*, and the Chow rings and Orlik–Solomon algebras of the local matroids are isomorphic.*

(3) $P_{X_1}(q) = 1 + 17q + 42q^2 + 42q^3 + 17q^4 + q^5$ *and* $P_{X_2}(q) = 1 + 12q + 47q^2 + 47q^3 + 12q^4 + q^5$*. In particular* $H^*_T(X_1) \not\cong H^*_T(X_2)$*, even as graded vector spaces.*

Proof. (3) Summing (2.1) over S_5 gives the polynomials. Independently, the GKM presentation of $H^*_T(X_i)$, computed in Macaulay2 as a graded $\mathbb{Q}[t_1, \ldots, t_5]$-module, has Hilbert series $P_{X_i}(q)/(1-q)^5$ with the same numerators. Since the actions are equivariantly formal, $\dim H^2_T(X_i) = \beta_2 + 5\beta_0$, which is 22 for X_1 and 17 for X_2.

(1) Smoothness, dimension and $H^{\mathrm{odd}} = 0$ were recalled in Section 2; connectedness follows from $\beta_0 = 1$. The weights at a fixed point are distinct roots $e_a - e_b$, hence pairwise independent. An element of T acting trivially lies in the common kernel of the weights $e_a - e_b$ over the edges of a connected graph, which is the diagonal circle, and the diagonal circle acts trivially on $\mathrm{Fl}(\mathbb{C}^5)$. So the effective torus has dimension 4 and the complexity is $5 - 4 = 1$.

(2) follows from Lemmas 4.1 and 4.2 and Theorem 2.1. □

Remark 4.4. A computer search over all pairs of Hessenberg functions with $n \leqslant 6$ finds smaller pairs in $\mathrm{Fl}(\mathbb{C}^4)$, for example $h = (2,3,3,4)$ and $(2,2,4,4)$, but these varieties are disconnected and β_0 merely counts components. Connected pairs also exist in $\mathrm{Fl}(\mathbb{C}^6)$, such as $h = (2,3,4,6,6,6)$ and $(2,3,5,5,6,6)$, with local matroid $U_{2,3} \oplus U_{1,1}^{\oplus 3}$ and Betti numbers $(1,43,179,274,179,43,1)$ and $(1,29,175,310,175,29,1)$.

The global face posets of X_1 and X_2 are not isomorphic: $\mathrm{Cl}_{\Gamma_{h_1}}$ and $\mathrm{Cl}_{\Gamma_{h_2}}$ have 200 and 190 elements of rank 2. This is no accident, as the next section shows.

5 The face poset determines the Betti numbers in type A

Question 1 ([16])**.** Let X_1, X_2 carry torus actions with $H^{\mathrm{odd}}(X_1) = H^{\mathrm{odd}}(X_2) = 0$. Does $S(X_1) \cong S(X_2)$ imply $\beta_j(X_1) = \beta_j(X_2)$ for all j?

For quasitoric manifolds the answer is yes, since $\beta_{2i} = h_i$ of the orbit polytope. The converse fails already there: a pentagonal prism and a cube with one vertex cut off have the same h-vector but different face lattices.

5.1 Rigidity of cluster-permutohedra

For $y \in \mathrm{Cl}_\Gamma$ let $\nu(y)$ be the number of minimal elements below y. An element $y = (C, A)$ lies above the minimal element given by a bijection $p\colon [n] \to [n]$ exactly when p maps every block onto its label set, so $\nu(y) = \prod_{V_i \in C} |V_i|!$. In particular Cl_Γ has $n!$ minimal elements.

Lemma 5.1. *Let $x \in \mathrm{Cl}_\Gamma$ be minimal. Then $(\mathrm{Cl}_\Gamma)_{\geqslant x} \cong \mathrm{Flats}(\Gamma)$, and its atoms correspond to the edges of Γ. The join of the atoms of a set S of edges has rank $n - c(S)$, where $c(S)$ is the number of components of $([n], S)$. For distinct edges e, f, the join of their atoms has $\nu = 6$ if e and f share a vertex and $\nu = 4$ otherwise. Three pairwise adjacent edges either form a triangle, and then the join of their atoms has rank 2, or share a vertex, and then it has rank 3.*

Proof. Above x the assignment is determined by the clustering, and the join of the atoms of S is the clustering into the components of $([n], S)$. For two edges this clustering has one block of size 3, or two blocks of size 2. Three pairwise adjacent edges span 3 vertices if they form a triangle and 4 otherwise. □

Call a set F of edges of a simple graph a *star family* if any two edges of F share a vertex and no three form a triangle. Let $E(v)$ be the set of edges at a vertex v.

Lemma 5.2. *(1) All edges of a star family with at least two edges contain a common vertex.*
(2) The maximal star families with at least two edges are exactly the sets $E(v)$ with $\deg v \geqslant 2$.

Proof. (1) Let uv and vw be in the family. A third edge meets both; if it avoids v, it is uw, and the three edges form a triangle. (2) If $\deg v \geqslant 2$, $E(v)$ is a star family. An edge $g \notin E(v)$ meeting every edge of $E(v)$ meets each at its other endpoint; as these are distinct, $E(v) = \{vu, vw\}$ and $g = uw$, a triangle. So $E(v)$ is maximal, and by (1) every maximal star family with at least two edges is of this form. □

Proposition 5.3. *Let $\varphi\colon E(\Gamma) \to E(\Gamma')$ be a bijection between the edge sets of simple graphs such that φ and φ^{-1} preserve adjacency of edges and send triangles to triangles. Then φ is induced by a bijection ψ between the non-isolated vertices: $\varphi(uv) = \psi(u)\psi(v)$ for every edge uv.*

Proof. By Lemma 5.2, $\varphi(E(v)) = E(\psi(v))$ defines a bijection ψ between the vertices of degree at least 2. Let uv be an edge. If $\deg u, \deg v \geqslant 2$, then $\varphi(uv) \in E(\psi(u)) \cap E(\psi(v))$, so $\varphi(uv) = \psi(u)\psi(v)$. If $\deg u \geqslant 2$ and $\deg v = 1$, then $\varphi(uv) = \psi(u)v'$; were $\deg v' \geqslant 2$, we would have $v' = \psi(w)$ with $uv \in E(w)$, so $w = v$, which is impossible. Put $\psi(v) = v'$, which is well defined

since v lies on one edge only. If $\deg u = \deg v = 1$, then uv is a component K_2, and by the previous cases applied to φ^{-1} so is $\varphi(uv)$; map u, v to its endpoints. The same construction for φ^{-1} gives the inverse of ψ. □

Theorem 5.4. *Let Γ and Γ' be simple graphs on $[n]$ and $[n']$, $n, n' \geqslant 1$. If $\mathrm{Cl}_\Gamma \cong \mathrm{Cl}_{\Gamma'}$, then $\Gamma \cong \Gamma'$.*

Proof. Let Φ be an isomorphism. Counting minimal elements gives $n! = n'!$, so $n = n'$. Fix a minimal x. Then Φ restricts to an isomorphism of lattices $(\mathrm{Cl}_\Gamma)_{\geqslant x} \to (\mathrm{Cl}_{\Gamma'})_{\geqslant \Phi(x)}$ and induces a bijection $\varphi\colon E(\Gamma) \to E(\Gamma')$ on atoms. Since Φ preserves joins in these intervals, rank and v, Lemma 5.1 shows that $\varphi^{\pm 1}$ preserve adjacency of edges and triangles. By Proposition 5.3, Γ and Γ' are isomorphic after removing isolated vertices, and both have n vertices. □

Remark 5.5. Adjacency alone is the line graph. Whitney's theorem [21] together with the rank of three edges, which separates K_3 from $K_{1,3}$, would also prove Theorem 5.4. For the partial case, where a partition of the vertices must be tracked as well, one would need the edge isomorphism to be induced by a vertex bijection; by Jung's theorem [15] this holds for connected graphs with more than four vertices but fails for $K_{1,3} + e$, $K_4 - e$ and K_4. Proposition 5.3 uses the triangles that the poset sees, and has no exceptions.

Corollary 5.6. *Let each of X and X' be either $M_{\Gamma,\lambda}$ with Γ connected and λ generic, or $\mathrm{Hess}(M, h)$ with Γ_h connected. If $S(X) \cong S(X')$, then $\beta_j(X) = \beta_j(X')$ for all j, and $H^*_T(X) \cong H^*_T(X')$ as graded rings. If X and X' are both Hessenberg varieties, this is an isomorphism of $H^*(BT)$-algebras; if both are isospectral manifolds, it is one up to an automorphism of T.*

Proof. By Theorem 2.3 the face posets are Cl_Γ and $\mathrm{Cl}_{\Gamma'}$ for the corresponding graphs, so $\Gamma \cong \Gamma'$ by Theorem 5.4. If π is a permutation with $\pi(\Gamma) = \Gamma'$ and P_π its permutation matrix, conjugation $A \mapsto P_\pi A P_\pi^{-1}$ preserves the spectrum and the zero pattern up to relabelling, $(P_\pi A P_\pi^{-1})_{\pi(i)\pi(j)} = a_{ij}$. It is therefore a diffeomorphism $M_{\Gamma,\lambda} \to M_{\Gamma',\lambda}$, equivariant along the automorphism $t \mapsto P_\pi t P_\pi^{-1}$ of T. This proves the corollary for two isospectral manifolds. A Hessenberg variety and its twin $M_{\Gamma_h,\lambda}$ have equal Betti numbers and equivariant cohomology rings isomorphic as graded rings (Theorem 2.3), which gives the general case. Finally, let $X = \mathrm{Hess}(M, h)$ and $X' = \mathrm{Hess}(M, h')$ with $\pi(\Gamma_h) = \Gamma_{h'}$. In the GKM description, $H^*_T(X)$ consists of the $(f_w)_{w \in S_n}$ with $f_w - f_{w \circ (i\,j)} \in (t_{w(i)} - t_{w(j)})$ for every edge ij of Γ_h. Since $(\pi(i)\,\pi(j)) \circ \pi = \pi \circ (i\,j)$, the map $(f_w) \mapsto (f'_{w'})$, $f'_{w'} = f_{w'\pi}$, sends these conditions to the conditions for $\Gamma_{h'}$, and it is $H^*(BT)$-linear. □

The corollary does not use $H^{\mathrm{odd}} = 0$. In particular it covers $M_{K_{1,3},\lambda}$ and the manifolds $M_{C_n,\lambda}$, $n \geqslant 4$, which are not equivariantly formal [4, 6].

5.2 Partial Hessenberg varieties

Let $X = \mathrm{Hess}_{\bar{k}}(M, h)$ and $G = \Gamma_h(\bar{k})$. For a partition $W = \{W_1, \ldots, W_l\}$ of $[n]$ let $T_W \subset T$ be the subtorus of the elements whose coordinates are equal within every block. The following description of the faces is [4, Thm. 2], in the form in which it appears in the proof given there. As Theorem 5.8 depends on its details, we include a proof.

Lemma 5.7. *The faces of X correspond to the families $\{(W_j, c_j)\}_{j=1}^{l}$, where $W_1, \ldots, W_l$ are the blocks of a partition of $[n]$, each $c_j \in \mathbb{Z}^s_{\geqslant 0}$ is a vector of part counts with $\sum_a c_j(a) = |W_j|$, $\sum_j c_j = \bar{k}$, and for every j either $|W_j| = 1$, or the support $\{a \mid c_j(a) > 0\}$ has at least two elements and induces a connected subgraph of Γ_h. The fixed points of the face are the J with $|J_a \cap W_j| = c_j(a)$ for all a and j, so their number is*

$$\nu\big(\{(W_j, c_j)\}\big) = \prod_j \frac{|W_j|!}{\prod_a c_j(a)!}.$$

The rank of the face is $n - l$, and $\{(W'_i, c'_i)\} \leqslant \{(W_j, c_j)\}$ if and only if every W'_i lies in some W_j and $c_j = \sum_{W'_i \subset W_j} c'_i$ for all j. For every fixed point J, $S(X)_{\geqslant J} \cong \mathrm{Flats}(G)$.

Proof. A subgroup $H \subset T$ has the same invariant subspaces in $\mathbb{C}^n$ as T_W, where W is the partition of $[n]$ into the classes on which the characters $t_x|_H$ coincide. Hence every stabilizer is some T_W, and the faces are the connected components Y of the sets X^{T_W} for which T_W is the stabilizer of a generic point of Y; such a face has rank $n - l$.

A T_W-invariant subspace is the direct sum of its intersections with the $\mathbb{C}^{W_j}$, so a flag in X^{T_W} is a direct sum of flags $V_\bullet^{(j)}$ in $\mathbb{C}^{W_j}$. Since M preserves $\mathbb{C}^{W_j}$, the sum satisfies the Hessenberg condition if and only if every summand does. The dimensions $c_j(a) = \dim V^{(j)}_{K_a} - \dim V^{(j)}_{K_{a-1}}$ are locally constant, and the flags with given c form the set

$$Y(W, c) \cong \prod_{j=1}^{l} \mathrm{Hess}_{c_j}(M|_{\mathbb{C}^{W_j}}, h),$$

where the parts a with $c_j(a) = 0$ are omitted, so that the jth factor is the partial Hessenberg variety of the induced subgraph of Γ_h on the support of c_j. Its fixed points are as stated. At a fixed point J, the tangent weights of $Y(W, c)$ are the weights $t_y - t_x$ of X with x and y in one block. They span a space of dimension $n - l$ if and only if every block induces a connected subgraph of G, the vertices of G being labelled through J; that is, if and only if every block is a single element or has counts supported on a connected set of at least two parts. If this holds, every factor is connected, by (2.2) or by Lemma 5.10, so $Y(W, c)$ is connected with generic stabilizer T_W and is a face. If it fails, the generic stabilizer of each component of $Y(W, c)$ is larger than T_W, and the component is counted for a finer partition.

Two faces $Y' \subset Y$ have generic stabilizers $T_{W'} \supset T_W$, so W' refines W, and a common fixed point shows that the counts add up; the converse holds because Y' is connected, is fixed by T_W and meets Y. The faces containing J correspond to the partitions W whose blocks induce connected subgraphs of G labelled through J, which are the flats of the graphic matroid of G, in accordance with Theorem 2.1. □

Theorem 5.8. *If X and X' are partial regular semisimple Hessenberg varieties with $S(X) \cong S(X')$, then $\beta_j(X) = \beta_j(X')$ for all j.*

The proof rests on three lemmas.

Lemma 5.9 (Splitting)**.** *Let $Q_1, \dots, Q_r$ be the connected components of Γ_h and P_{Q_i} the Poincaré polynomial of the partial Hessenberg variety of the restriction of $(\bar{k}, h)$ to Q_i. Then*

$$P_X(q) = \#X^T \cdot \prod_{i=1}^{r} \frac{P_{Q_i}(q)}{P_{Q_i}(1)},$$

and a component consisting of a single part contributes the factor 1.

Proof. The components of the indifference graph Γ_h are intervals of parts, and every pair counted in (2.2) lies in one component. Put $U_i = \bigcup_{a \in Q_i} J_a$. The pairs counted inside Q_i depend only on the relative order of the elements of U_i, that is, on a fixed point of the variety of Q_i. Summing over the choices of $U_1, \dots, U_r$ gives $P_X = \binom{n}{|U_1|,\dots,|U_r|} \prod_i P_{Q_i}$, and setting $q = 1$ gives $\#X^T = \binom{n}{|U_1|,\dots,|U_r|} \prod_i P_{Q_i}(1)$. □

Lemma 5.10 (Fibration)**.** *Put $\tilde{h}(i) = K_{h(a)}$ for $K_{a-1} < i \leqslant K_a$, and let $\pi\colon \mathrm{Fl}(\mathbb{C}^n) \to \mathrm{Fl}_{\bar{k}}$ be the projection. Then $\pi^{-1}(X) = \mathrm{Hess}(M, \tilde{h})$, the restriction of π is a fibre bundle with fibre $\prod_a \mathrm{Fl}(\mathbb{C}^{k_a})$, and*

$$P_{\mathrm{Hess}(M,\tilde{h})}(q) = P_X(q) \cdot \prod_{a=1}^{s} [k_a]_q!.$$

Moreover, $\Gamma_{\tilde{h}}$ is the inflation $\Gamma_h(\bar{k})$ with every part made a clique.

Proof. A full flag $F_\bullet$ lies in $\mathrm{Hess}(M,\tilde{h})$ iff $MF_i \subset F_{\tilde{h}(i)}$ for all i. For i in part a we have $F_i \subset F_{K_a}$ and $\tilde{h}(i) = \tilde{h}(K_a)$, so the condition at i follows from the condition at K_a, and the conditions at $K_1, \dots, K_s$ say that $\pi(F_\bullet) \in X$. The projection π is a holomorphic bundle with fibre $\prod_a \mathrm{Fl}(V_{K_a}/V_{K_{a-1}})$ over $V_\bullet$, so it restricts to a bundle over X. In particular X is smooth, since $\mathrm{Hess}(M,\tilde{h})$ is smooth and π is a submersion. The cohomology of the fibre is generated by Chern classes of tautological bundles defined on the total space, and all spaces involved have only even cohomology, so the Leray–Hirsch theorem gives the product formula, with $P_{\mathrm{Fl}(\mathbb{C}^k)} = [k]_q!$. Finally, $i < j$ is an edge of $\Gamma_{\tilde{h}}$ iff $j \leqslant K_{h(a)}$ for the part a of i, that is, iff j lies in part a or in a part b with $a < b \leqslant h(a)$. □

Lemma 5.11 (Reconstruction)**.** *Let $G = \Gamma_h(\bar{k})$ with its partition into parts. The poset $S(X)$ determines $\#X^T$ and the multiset of isomorphism classes, as graphs with a vertex partition, of the components of G that have an edge.*

Proof. The number $\#X^T$ is the number of minimal elements. Fix a fixed point J and identify $S(X)_{\geqslant J}$ with $\mathrm{Flats}(G)$ as in Lemma 5.7. For edges uv and vw the join of their atoms has the single nontrivial block $\{u,v,w\}$, so by the formula for ν in that lemma $\nu = 3$ if u and w lie in one part and $\nu = 6$ otherwise, since adjacent vertices lie in different parts. Two disjoint edges give $\nu = 4$. As in the proof of Theorem 5.4, an isomorphism $\Phi\colon S(X) \to S(X')$ induces a bijection φ between the edge sets of the inflations that preserves the rank and ν of the join of every set of edges. By the values of ν, $\varphi^{\pm 1}$ preserve adjacency, and since the rank statement of Lemma 5.1 holds verbatim for $\mathrm{Flats}(G)$, they preserve triangles. By Proposition 5.3, φ is induced by a bijection ψ between the non-isolated vertices.

The map ψ preserves parts. Two distinct non-isolated vertices u, w lie in one part iff they have a common neighbour v with $\nu(a_{uv} \vee a_{vw}) = 3$: a non-isolated vertex is a copy of a part a of a component Q of Γ_h with at least two parts, so the copies of a have a common neighbour, and $\nu = 3$ forces u, w into one part. Since ψ preserves adjacency and φ preserves ν, ψ preserves this relation. Finally, the components of G with an edge are the inflations G_Q of the components Q of Γ_h with at least two parts, and copies of the remaining parts are isolated, so ψ matches these components as graphs with parts. □

Proof of Theorem 5.8. By Lemma 5.9 it suffices to show that $S(X)$ determines $\#X^T$ and $\prod_Q P_Q(q)/P_Q(1)$ over the components Q of Γ_h with at least two parts. For such Q, Lemma 5.10 gives $P_Q = P_{\mathrm{Hess}(M_Q,\tilde{h}_Q)}/\prod_{a\in Q}[k_a]_q!$. The graph $\Gamma_{\tilde{h}_Q}$ is connected and is obtained from G_Q by making the parts cliques, and isomorphic connected graphs $\Gamma_{\tilde{h}}$ give Hessenberg varieties with equal Betti numbers (the twins are then equivariantly diffeomorphic, as in the proof of Corollary 5.6). Hence $P_{\mathrm{Hess}(M_Q,\tilde{h}_Q)}$ depends only on the isomorphism class of G_Q with its parts; so do the part sizes. Lemma 5.11 completes the proof. □

5.3 Remarks

Theorem 5.8 includes partial flag varieties ($\Gamma_h = K_s$) and Grassmannians, where it is classical, and the full case $\bar{k} = (1, \dots, 1)$. For disconnected Γ_h the variety $\mathrm{Hess}(M,h)$ is a disjoint union of products of smaller Hessenberg varieties, and Theorem 5.8 covers it as well.

The statements of this subsection were rechecked by a program written independently of the proofs: Lemma 5.7 against the faces computed from the GKM graph for all 186 pairs $(\bar{k},h)$ with $n \leqslant 5$; formula (2.2) against the number of $\mathbb{F}_q$-points of $\mathrm{Hess}_{\bar{k}}(M,h)$ for $n = 4$ and $q = 5, 7$; and the statement that the data recovered in Lemma 5.11 determine the Poincaré polynomial, for all pairs with $n \leqslant 7$. Lemma 5.10 and Theorem 5.4 were also checked by computer: the identity of Lemma 5.10 for all 2 949 pairs $(\bar{k},h)$ with $n \leqslant 7$, and Theorem 5.4 by exact poset isomorphism for all 208 graphs with $n \leqslant 6$.

6 Open questions

(1) *Other Lie types.* Regular semisimple Hessenberg varieties exist for every reductive group. Is there an analogue of Theorem 5.4 for their face posets? Outside type A the local weight matroids are no longer graphic, and the star argument of Proposition 5.3 does not apply directly. The face poset has the same shape in every type: its elements are the pairs (F, wW_F) of a flat F of the matroid of the ideal $I \subset \Phi^+$ defining the variety and a coset of the reflection subgroup W_F generated by F. A computation for all $1\,230$ ideals of the root systems of rank at most 4 and of A_5, B_5, C_5, D_5 found no counterexample: the number of fixed points and the lattice of flats of I with the function $F \mapsto |W_F|$, both read off from the face poset, determine the Betti numbers, and for connected varieties they determine I up to diagram automorphisms and the duality $\alpha \leftrightarrow \alpha^\vee$.

(2) *A formula.* In complexity zero, $S(X)$ determines the Betti numbers through the h-vector. Is there a formula for β_j in terms of $S(X)$ for Hessenberg varieties, as opposed to the reconstruction argument given here?

(3) *Beyond algebraic actions.* Question 1 remains open for general equivariantly formal torus manifolds of positive complexity.

(4) *Oriented weights.* If X carries an invariant stably complex structure, the weights have signs and $S(X)_{\geqslant p}$ refines to an oriented matroid. What does the resulting oriented structure determine? For flag varieties and regular semisimple Hessenberg varieties with their full torus it determines nothing new: the signed weights at the fixed point w are the image under w of those at the identity, so the oriented matroid is the same at every fixed point. For subtori it does carry information; for a generic circle the oriented matroid at p is the pair $\{\mathrm{ind}(p), n - \mathrm{ind}(p)\}$, and these pairs give the Betti numbers.

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